



Global Retail Sales Dashboard

Why Analyze Sales, Returns, Profitability, and Regional Performance?

You are working as a Data Analyst at GlobalSuper Sales, a global retail company. The company operates across multiple regions, markets, and product categories (Furniture, Office Supplies, Technology).

Business Problem

Management has observed:

- Profitability is inconsistent across regions
- High discounts are sometimes reducing profits
- Some orders are being returned, negatively impacting revenue
- Sales targets vary by region and leadership

Impact of Analysis

A comprehensive analysis in Power BI will enable leadership to:

- Identify key profit drivers
- Reduce losses due to returns and discounts

- Evaluate regional performance efficiently
 - Make data-driven strategic decisions
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Your Role

1. Understand the Data Provided

Global Retail Sales Dataset → [Click here to get Dataset](#)

Datasets Overview

- **Orders Data**
 - Fields: Order ID, Order Date, Ship Date, Ship Mode, Customer Details, Segment, Market, Region, Product Details, Sales, Quantity, Discount, Profit, Shipping Cost, Priority
 - **Returns Data**
 - Fields: Order ID, Market, Returned (Yes/No)
 - **People Data**
 - Fields: Person, Region (Regional Sales Manager assignment)
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2. Clean and Transform the Data

Key Tasks

- Standardize date formats (Order Date, Ship Date)
- Clean Sales/Profit columns (remove currency symbols, ensure numeric type)

- Handle missing values (e.g., missing postal codes)
- Remove duplicate orders, if any
- Create calculated columns as needed (e.g., Profit Margin = Profit / Sales)
- Transform Returns data into a usable Yes/No flag linked to Orders

3. Build a Relational Data Model

Relationships

- Orders ↔ Returns (via Order ID)
- Orders ↔ People (via Region)

Modeling Guidance

- Orders = Fact table
- Returns, People = Dimension tables
- Star schema is recommended for optimized reporting

4. Create DAX Measures to Answer Key Business Questions

Business Questions

- Calculate Total Sales, Total Profit, and Total Quantity
- Determine Profit Margin % at order level
- Calculate Sales by Segment and Region
- Compare Actual Sales vs. Target (simulate or define target measure)
- Identify Top 10 Products by Sales

- Calculate Return Rate % ($\text{Returned Orders} \div \text{Total Orders}$)
- Calculate Year-over-Year Sales Growth %
- Identify High-Value Customers (Top 10% revenue contributors)
- Calculate Average Shipping Cost per Region
- Create a Monthly Sales Trend using time intelligence

5. Dashboard Development

Expected KPIs

- Total Sales
- Total Profit
- Profit Margin %
- Sales vs Target %
- Top 10 Products by Sales
- Return Rate %
- Sales Trend (Monthly / Yearly)
- Regional Performance (Sales, Profit, Return Rate)
- Segment-wise Contribution (Consumer, Corporate, Home Office)

Business Questions to Answer via Dashboard

- What are the overall sales and profit trends over time?
- Which regions are performing well, and which are underperforming?
- Which product categories and sub-categories are driving the most profit?

- What is the impact of discounts on profit margins?
- How many orders are being returned, and which regions have the highest return rates?
- Which customers contribute the most revenue and profit?
- Which shipping modes and priorities are most profitable?
- Are sales managers achieving balanced performance across their regions?

Must-Include Dashboard Features

- Drill-through (Region → Product → Customer)
- Conditional Formatting (highlight negative profits in red)
- Sales vs Target visual (bar, gauge, or KPI card)
- Interactive Filters/Slicers (Year, Region, Segment, Category)
- KPI Cards (Total Sales, Profit, Profit Margin %, Return Rate, YoY Growth)

6. Provide Actionable Insights and Recommendations

Questions to Explore

- Why is one region underperforming despite high sales?
- Are high discounts helping sales growth or reducing profits?
- Which customer segment is most profitable and should be prioritized?
- Do certain shipping modes drive higher costs and lower profitability?
- What patterns exist in returned orders, and how can returns be reduced?
- Which products or categories should the company expand or drop?
- Are regional sales managers driving balanced results?
- What strategies can management apply to improve profit margins?

Expected Deliverables

- Cleaned and transformed Power BI Data Model
- DAX measures answering the business challenge questions
- Interactive Power BI Dashboard with KPIs and required features
- Business Insights Report (1–2 pages of recommendations for management)